Portaldot Hacker House 2026 · Multi-Chain & Cross-Chain Infra

# Kurs

Price feeds for Portaldot V3, signed by independent reporters, readable from Solidity and ink!, portable to any EVM chain.

Kurs: Indonesian for exchange rate, the number on every money-changer board.

# Portaldot already has money apps. None of them can read a price.

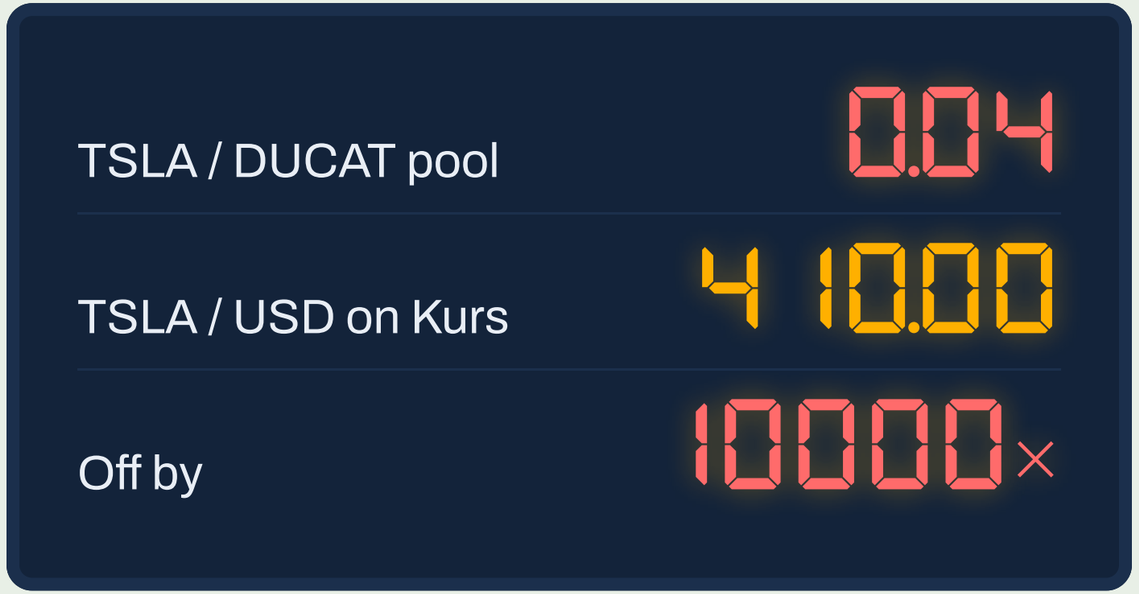
| Checked on V3 testnet, spec 3002 | Found |
|---|---|
| Oracle pallet (index 85) | **0 values** |
| Oracle operator set | **empty** |
| Oracle precompile for EVM | **none** |
| Chainlink, Pyth, RedStone | **not deployed** |
| Tokenised stocks on the native DEX | TSLA, NVDA, AAPL, AMZN, META |
| Hacker-house lending, stablecoin, escrow apps | a dozen and counting |

# Ask the pools what TSLA is worth

Route one TSLA token through Portaldot's own pools (TSLA → DUCAT → tPOT → pUSD) and you get **about $0.04**.

Kurs says **about $410**.

A lending market that prices collateral from these pools can be drained by anyone with a few swaps. That is how most DeFi exploits start.



# How Kurs works

1

### Read

Reporters read exchanges, aggregators, Chainlink on Ethereum, FX and equity quotes, and Portaldot's own pools.

2

### Sign

Each reporter signs its own number with EIP-712. No chain id in the domain.

3

### Relay

Anyone submits the bundle. The sender is not trusted; the signatures are.

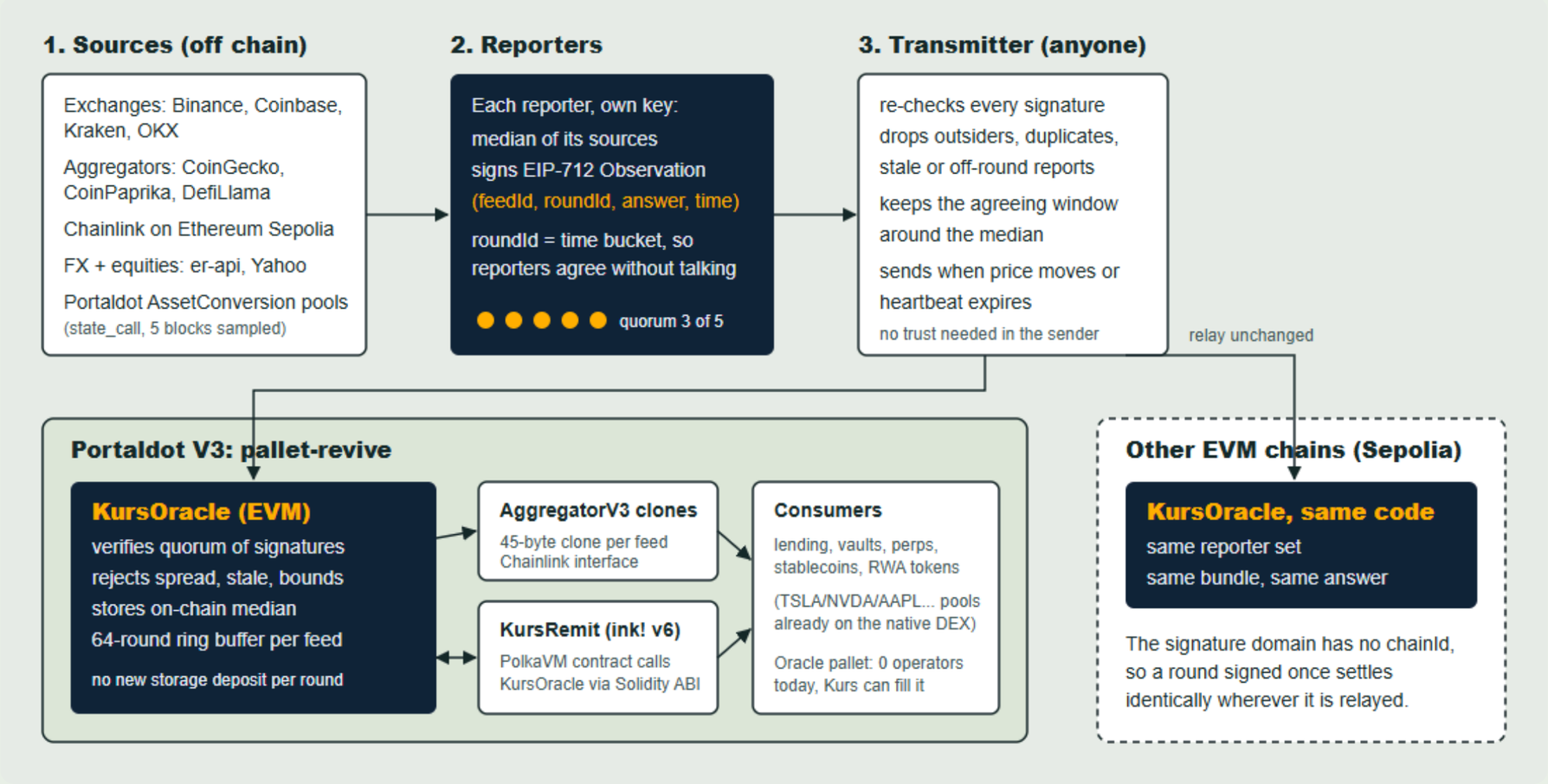
4

### Settle

KursOracle checks a majority quorum, freshness, spread and bounds, and stores the median.

A minority of dishonest reporters cannot move the median, and cannot forge a round.

# One oracle, both VMs, every chain



# Why this has to be Portaldot

**Dual VM in one pallet.** Solidity reads Kurs through the Chainlink interface. ink! v6 reads the same contract through a Solidity-ABI call. One oracle, not two.

**A native price to export.** tPOT/pUSD comes from `pallet-asset-conversion` through a runtime API, sampled over 20 minutes of blocks, then relayed to Ethereum with the same signatures.

**A slot already waiting.** The runtime ships an Oracle pallet with no operators. Kurs reporters are ready to fill it, so native pallets and contracts read the same prices.

**Built for Revive's costs.** A 64-round ring buffer and 45-byte clones mean a new round adds no storage deposit.

# Built during the hacker house

11

feeds: BTC, ETH, DOT, USDT, USD/IDR, five equities, tPOT

73

tests: 30 Solidity incl. fuzz and invariants, 17 TypeScript incl. anvil e2e, 26 Playwright incl. accessibility and layout shift; Slither in CI

2

VMs reading one oracle: EVM and PolkaVM (ink! v6)

5

reporters, quorum 3, median on chain

0

servers: the dashboard reads public RPCs in the browser

2

ways in: pushed by the relay, or pulled into your own transaction with ERC-7412

# Proof on chain

| What | Where |
|---|---|
| KursOracle v2, factory, 11 feed clones, KursConverter | Ethereum Sepolia, oracle `0xd3f6…cbc3`, all verified on Sourcify |
| Rounds pushed by 5 reporters, every signer on chain | RoundUpdated events, re-verified in the browser by the dashboard |
| ERC-7412 pull in one transaction | Multicall3: fulfillOracleQuery + convert, tx `0x9e2a…405f` |
| ink! v6 reads the Solidity oracle | pallet-revive node: KursRemit.latest() equals the oracle round |
| Portaldot-native price on Ethereum | tPOT/pUSD from AssetConversion, live on Sepolia |
| Live on Portaldot V3 | KursOracle `0x5bdd…56f4`, 11 feeds, rounds every ~64 min at ~0.1 tPOT, mirrored to Sepolia |
| Owned by a native Portaldot multisig | 2-of-3 pallet-multisig; acceptOwnership executed via asMulti + revive.call |
| ERC-7412 pull on Portaldot | fulfillOracleQuery then convert, $100 to Rp1,814,489.94 |

Every address and transaction hash:

`github.com/PugarHuda/kurs/blob/main/docs/evidence.md` ·

live: `kurs-ten-zeta.vercel.app`

# Who pays, and why

**Users:** every Portaldot app that moves value against a price. Lending (DotProof), stablecoins (Ballast, StableTap, pUSD), tokenised stocks (PASTX pools), escrow and payroll in IDR.

**Feed sponsorship:** an app pays a monthly POT fee per feed it needs at a given heartbeat. Sponsors fund reporter rewards and transmit gas.

**Export fees:** Portaldot-native prices (tPOT, ecosystem tokens) relayed to Ethereum, BNB and Base for bridges and CEX listings.

**Reporter market:** community operators stake POT to join the set and earn from sponsorship. Misbehaviour is visible on chain in every round's signer bitmap.

# Next

- Independent reporter operators from the Portaldot community
- Owner key behind Portaldot's native Multisig + Proxy with a delay
- Kurs reporters as operators of the runtime Oracle pallet
- Read precompile for the Oracle pallet
- Relay to Base, BSC, Arbitrum; POT feeds via bridge.portaldot.dev
- Paid equity data before mainnet; external audit

# Team

## Pugar Huda Mantoro

Built pdk, the Portaldot Dev Kit: 1st place, Portaldot Online Mini Hackathon S1, Builder Tools. Published on PyPI and npm.

Kurs is open source under MIT:
github.com/PugarHuda/kurs

